OPERATIONAL RESEARCH & OPTIMISATION

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Today’s lecture will cover

- Sensitive analysis
  - The interpretation of sensitive analysis
  - Use of sensitive analysis
- Network modelling
  - Terminology of network;
  - Transportation problem;
  - Assignment problem;
  - Minimum cost flow problems;
  - Maximum flow problem;
  - Shortest-path problem;

Sensitive analysis of LP

- Sensitive analysis is used to determine how the optimal solution is affected by changes, within specific ranges in:
  - The objective function coefficients c_j;
  - The right-hand side(RHS) values b_i.
- Sensitive analysis is important to the manager who must operate in a dynamic environment with imprecise estimations of parameters.
- Sensitive analysis allows him to ask certain what-if questions about the problem.

Sensitive analysis

$$
\begin{array}{ll}
\text{Maximise :} & Z = c_1x_1 + c_2x_2 + \ldots c_nx_n \\
\text{Subject to :} & a_{11}x_1 + a_{12}x_2 + \ldots a_{1n}x_n \le b_1 \\
& a_{21}x_1 + a_{22}x_2 + \ldots a_{2n}x_n \le b_2 \\
& \vdots \qquad\qquad\qquad\qquad \vdots \\
& a_{m1}x_1 + a_{m2}x_2 + \ldots a_{mn}x_n \le b_m \\
& x_1 \ge 0, x_2 \ge 0 \ldots x_n \ge 0
\end{array}
$$

- Greater attention is given to performing sensitivity analysis on the $\boldsymbol{b_i}$ and $\boldsymbol{c_j}$ parameters than on the $\boldsymbol{a_{ij}}$ parameters. Because:
- --On real problems with hundreds or thousands of constraints and variables, the effect of changing one $\boldsymbol{a_{ij}}$ value is usually negligible, but changing one $\boldsymbol{b_i}$ or $\boldsymbol{c_j}$ value can have real impact.
- --Furthermore, in many cases, the $\boldsymbol{a_{ij}}$ values are determined by the technology being used (the $\boldsymbol{a_{ij}}$ values are sometimes called technological coefficients), so there may be relatively little (or no) uncertainty about their final values.

- To understand the sensitive analysis

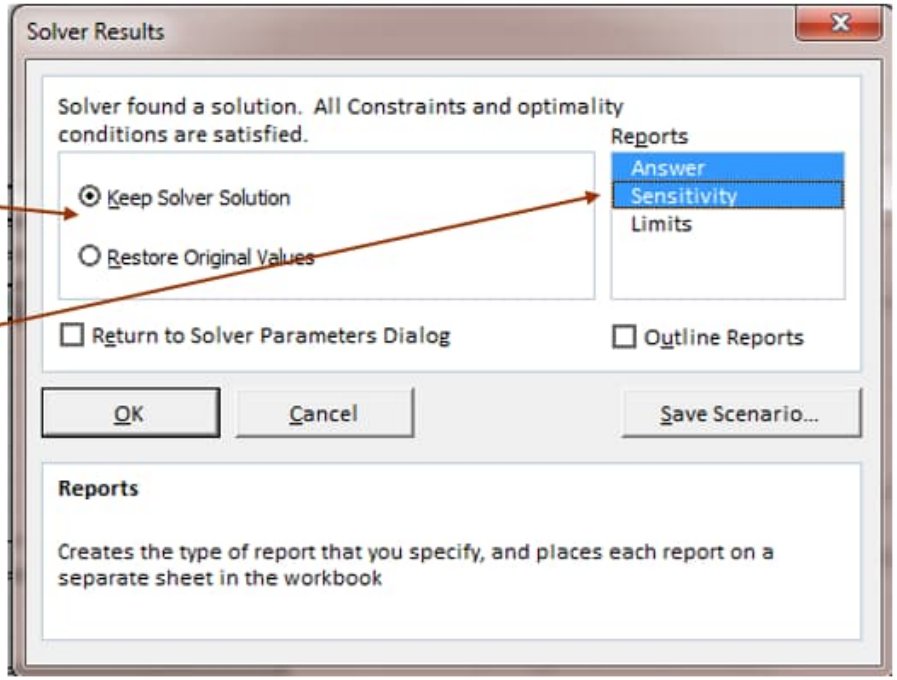
  We can use graphic method to visualize it;

  +

  Use EXCEL to generate the sensitive report.

Get sensitivity report in Excel

- Tick one of the two options
- Then select "Answer" and "Sensitivity reports
- Click OK. Answer and Sensitivity Reports will be generated in separate sheets



How to interpret the sensitivity report

- The top section of the report is for sensitivity to changes in the coefficients of the decision variables in the objective—topic 1
- The bottom section of the report is for sensitivity to changes in the right sides of the constraints—topic 2

Example Post-optimality analysis report in the Excel Solver.

Microsoft Excel 14.0 Sensitivity Report
Worksheet: [ORModels1-PM.xlsx]Wenbu
Report Created: 01/11/2014 13:17:34

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
H5	Hotdogs Produced	60	0	25	1E+30	17.85714286
H6	Buns Produced	42.85714286	0	10	25	10

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
C8	Used Labour	600	1.428571429	600	260	300
D8	Used Flour	428.5714286	0	800	1E+30	371.4285714
E8	Used Pork	1500	0.714285714	1500	1500	1300

The Excel Solver provides a Sensitivity Analysis report covering the objective function (upper table) and the constraints (lower table).

Example 1

TABLE 3.1 Data for the Wyndor Glass Co. problem

Plant	Production Time per Batch, Hours: Product 1	Production Time per Batch, Hours: Product 2	Production Time Available per Week, Hours
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	$3,000	$5,000	

Production time

Formulation as a Linear Programming Problem

The definition of the problem given above indicates that the decisions to be made are the number of batches of the respective products to be produced per week so as to maximize their total profit. Therefore, to formulate the mathematical (linear programming) model for this problem, let

x_1 = number of batches of product 1 produced per week

x_2 = number of batches of product 2 produced per week

Z = total profit per week (in thousands of dollars) from producing these two products

Thus, x_1 and x_2 are the *decision variables* for the model. Using the bottom row of Table 3.1, we obtain

$$Z = 3x_1 + 5x_2.$$

Example 1-continue

$$\text{Maximize} \quad Z = \boxed{3x_1 + 5x_2,}$$

They've used 3 and 5 instead of 3000 and 5000

subject to the restrictions

$$\begin{aligned} x_1 \quad\quad &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned}$$

and

$$x_1 \geq 0, \quad x_2 \geq 0.$$

Example 1-continue

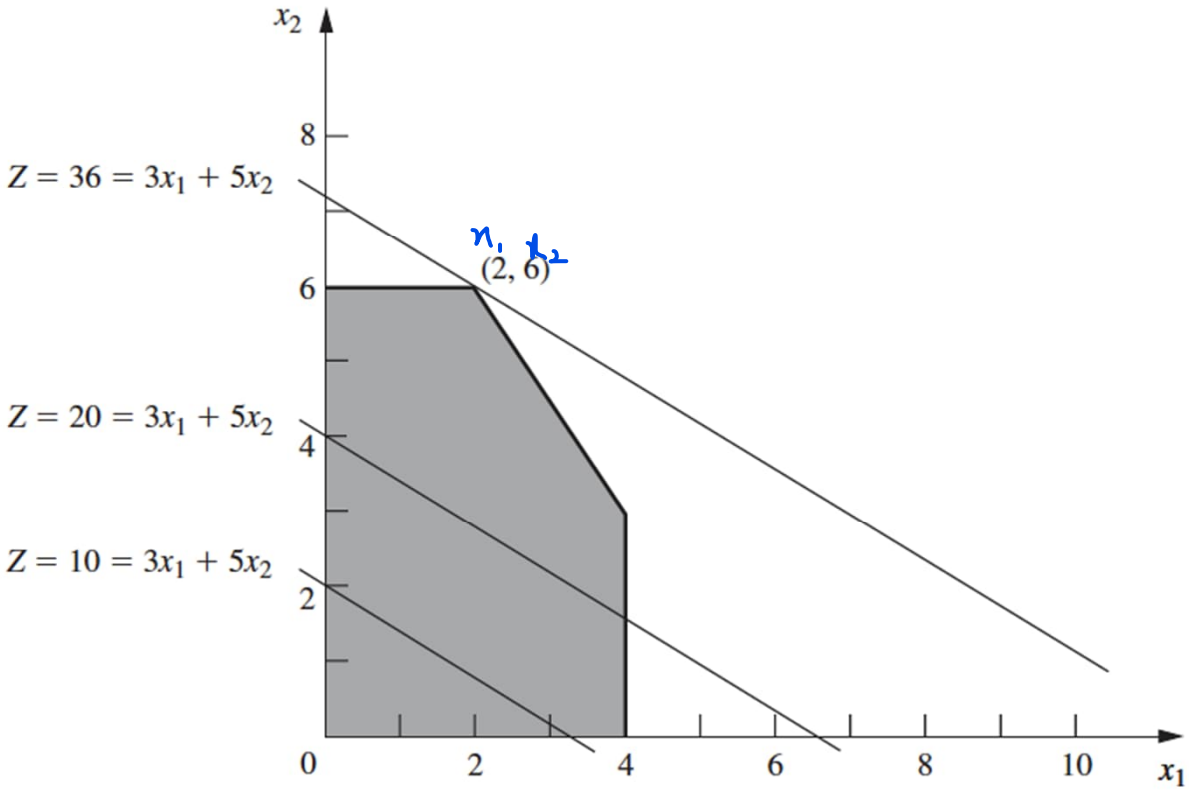


FIGURE 3.3
The value of (x_1, x_2) that maximizes $3x_1 + 5x_2$ is (2, 6).

Topic 1- Change in the objective function coefficients

- Let us consider how changes in the objective function coefficients might affect the optimal solution.
- The range of optimality for each coefficient provide the range of values over which the current solution will remain optimal.
- Managers should focus on those objective coefficients that have a narrow range of optimality and coefficients near the endpoints of the range.

Topic 1- Change in the objective function coefficients-EXCEL sensitive report

min < = c1<=max=0 <c1<=7500
2000 <= c2<= unlimited
c1 can change from 3000 to 7500

Reduced cost: if we increase the profit

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
C12	Batches Produced Doors	2	0	C1 3	4.5	3
D12	Batches Produced Windows	6	0	C2= 5	1E+30	3

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
E7	Plant 1 Used	2	0	4	1E+30	2
E8	Plant 2 Used	12	1.5	12	6	6
E9	Plant 3 Used	18	1	18	6	6

FIGURE 4.10
The sensitivity report provided by the Excel Solver for the Wyndor Glass Co. problem.

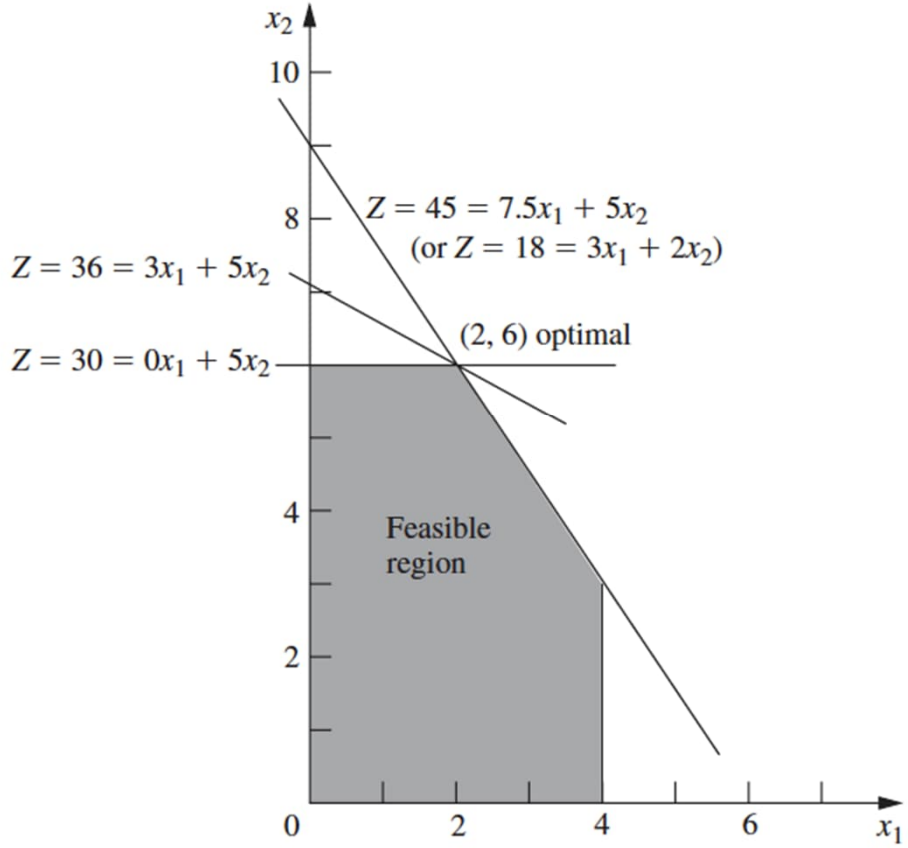
- Thus, 3-3<=c_1<=3+4.5; 0<= c_1 <=7.5
- ~~3~~ 5-3 <=c_2; ~~0~~ 2<= c_2

Topic 1- Change in the objective function coefficients

- Graphically, the limits of a range of optimality are found by **changing the slope of the objective function line** within the limits of the slopes of the binding constraint lines.

Topic 1- Change in the objective function coefficients-Graphic visualization

FIGURE 4.9
This graph demonstrates the sensitivity analysis of c_1 and c_2 for the Wyndor Glass Co. problem. Starting with the original objective function line [where $c_1 = 3$, $c_2 = 5$, and the optimal solution is (2, 6)], the other two lines show the extremes of how much the slope of the objective function line can change and still retain (2, 6) as an optimal solution. Thus, with $c_2 = 5$, the allowable range for c_1 is $0 \le c_1 \le 7.5$. With $c_1 = 3$, the allowable range for c_2 is $c_2 \ge 2$.



Reduced cost

- The reduce cost for **a decision variable whose value is 0** in the optimal solution is the amount the **variable's objective function coefficient would have to improve** before this variable could take a positive value.
  - Example: maximising the profits:
    - One of the decision variable Xa (number of product A)=0 and its associated reduced cost=-2.
    - This means the profit per unit on product A would need to increase by 2 before it would be profitable to produce any of product A.
- The reduced cost for a decision variable with a positive value is **0**.

Topic 1- Change in the objective function coefficients-EXCEL sensitive report

6 <=b1<=18

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
C12	Batches Produced Doors	2	0	3	4.5	3
D12	Batches Produced Windows	6	0	5	1E+30	3

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
E7	Plant 1 Used	2	0	4	1E+30	2
E8	Plant 2 Used	12	1.5	12	6	6
E9	Plant 3 Used	18	1	18	6	6

optimum solution utilisation (hours)

b values

The range it can vary to remain in the optimal solution

FIGURE 4.10
The sensitivity report provided by the Excel Solver for the Wyndor Glass Co. problem.

shadow price is the change in profit when the right-hand side co- efficient increased by one unit
eg: if you increase the R.H.S from 4 to 5, then the shadow price would not increase.
if we increased it from 12 to 13, the shadow price would increase by 1500

Topic 2-Change in the right-hand side of a constraint

- The term **shadow price** indicates the change in the optimal value of the objective when the **right side of some constraint** changes by one unit.
- If a resource constraint is binding in the optimal solution, the company is willing to pay up to some amount, the shadow price, to obtain more of the resource.

Topic 2-Change in the right-hand side of a constraint-EXCEL sensitive report

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
C12	Batches Produced Doors	2	0	3	4.5	3
D12	Batches Produced Windows	6	0	5	1E+30	3

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
E7	Plant 1 Used	2	0	4	1E+30	2
E8	Plant 2 Used	12	1.5	12	6	6
E9	Plant 3 Used	18	1	18	6	6

FIGURE 4.10
The sensitivity report provided by the Excel Solver for the Wyndor Glass Co. problem.

Topic 2-Change in the right-hand side of a constraint-Graphic visualization

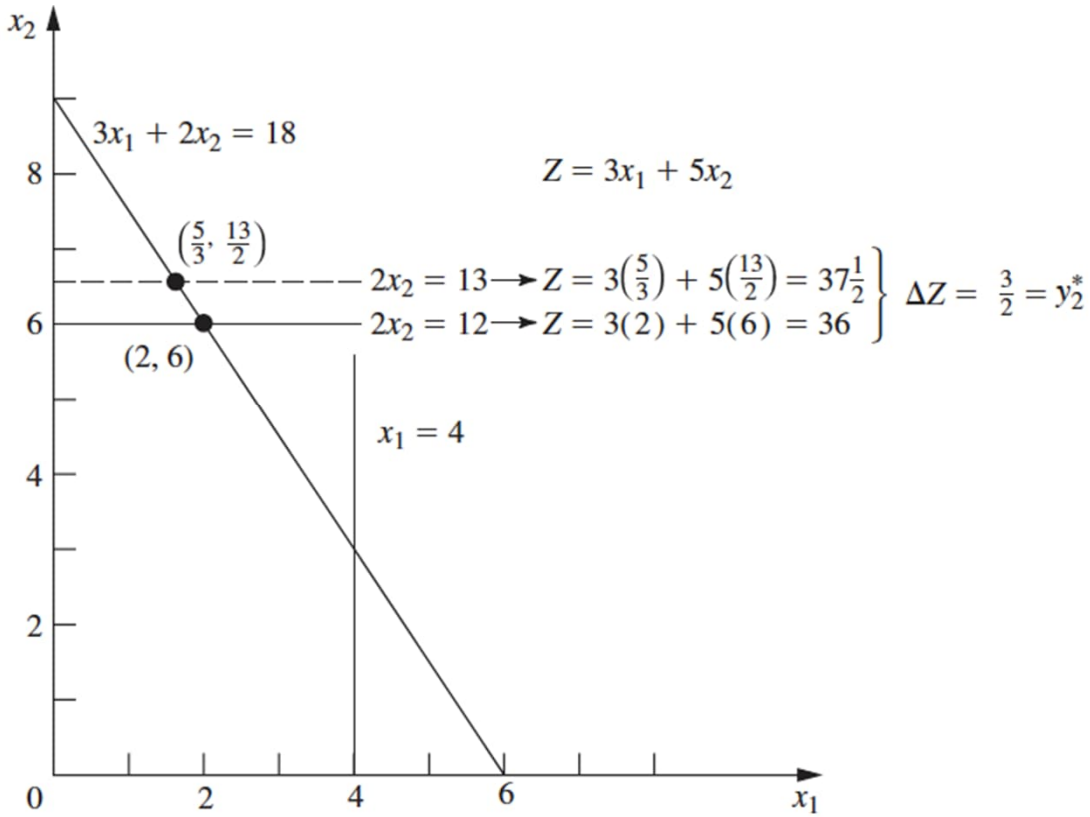


FIGURE 4.8
This graph shows that the shadow price is $y_2^* = \frac{3}{2}$ for resource 2 for the Wyndor Glass Co. problem. The two dots are the optimal solutions for $b_2 = 12$ or $b_2 = 13$, and plugging these solutions into the objective function reveals that increasing b_2 by 1 increases Z by $y_2^* = \frac{3}{2}$.

Use of post-optimiality analysis

- It can help answer:
  - How changes in the availably of constrained resources affect the optimal solutions and the objective function values?
  - What is the slack, if any, of resources in the optimal solution?
  - What resources are more critical in the optimisation problem?
  - Is there an alternative plan with the same profit of cost?
  - How can the current optimal solution be improved?
  - What cost is acceptable for increase resources?

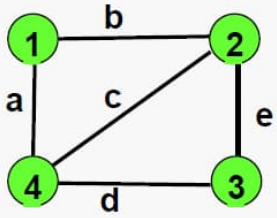
Part 2 Network Optimisation models

- Is a special type of Linear Programming involving networks;
- Efficient algorithms can achieve dramatic computational savings by exploring this special structure of the problems;
- It is important to become sufficiently familiar with these special types of problems that you can recognize them when they arise and apply the proper computational procedure.

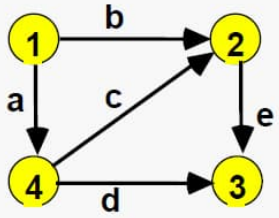
The terminology of networks

Node, Arc, Directed arc, A directed network (undirected network

Directed and Undirected Networks



An Undirected Graph



A Directed Graph

- Networks are used to transport commodities
  - physical goods (products, liquids)
  - communication
  - electricity, etc.
- The field of Network Optimization concerns optimization problems on networks

The terminology of networks

- Flow:

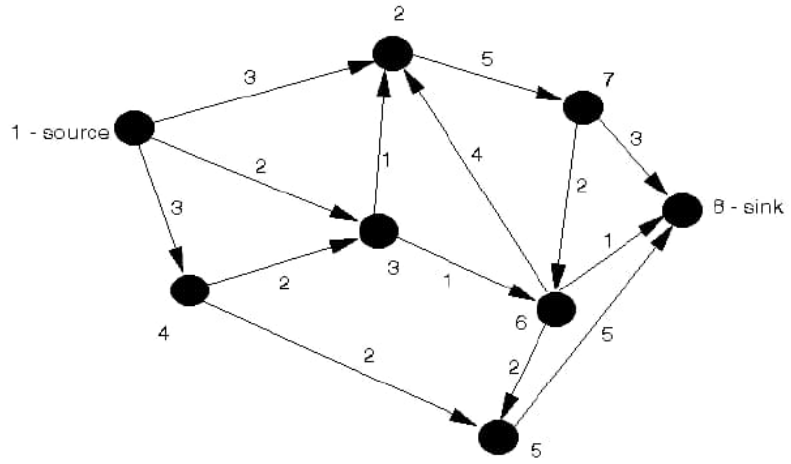
TABLE Components of typical networks

Nodes	Arcs	Flow
Intersections	Roads	Vehicles
Airports	Air lanes	Aircraft
Switching points	Wires, channels	Messages
Pumping stations	Pipes	Fluids
Work centers	Materials-handling routes	Jobs

Applications of Network Optimization

Network flow optimization problems arise in many real-world applications including:

- Communications
- Transportation
- Distribution
- Project planning
- Circuit design
- Production
- Financial planning



http://people.brunel.ac.uk/~mastjjb/jeb/or/netflow.html

Many network flow optimization problems (although not all) can be formulated as Linear Programming models.

Examples of network flow optimization problems:

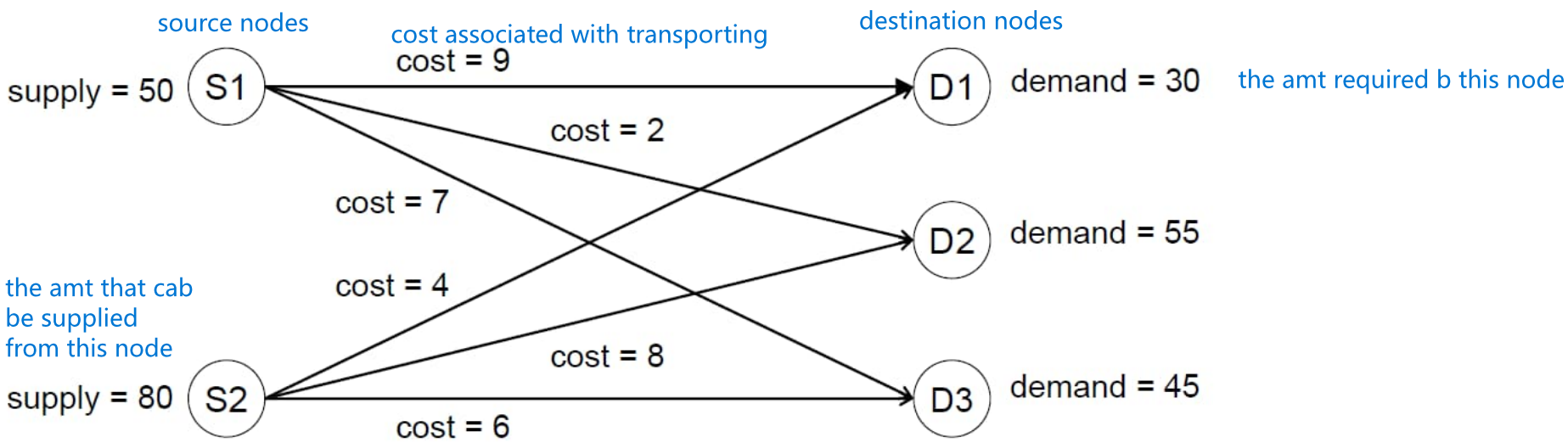
- Find the shortest route between two points in a city
- Find the critical path of activities in a large project
- Find the fastest route to send data through a network
- Design the best network for mobile communications
- Design the best layout for an electronic circuit
- Design the best route map for an airline
- Design the best evacuation plan for a city
- Find the maximum cash flow in a financial network
- Find the maximum flow in a water distribution network
- Find the minimum distribution cost in a supply chain

See attached file on BlackBoard: Applications of Networks

Transportation problem-a prototype example

Transportation Problem

Consider a directed network with a set of supply nodes (sources) and a set of demand nodes (destinations). A given transportation cost and unlimited flow capacity are associated to each arc. The objective is to distribute the entire commodity supply from the sources to the destinations such as to minimise the total transportation cost. Linear Programming can solve the transportation problem very efficiently.



Assumption of the transportation model

- The requirement assumption:
  - Each source has a fixed *supply* of units, where this entire supply must be distributed to the destinations. Similarly, each destination has a fixed *demand* for units, where this entire demand must be received from the sources.
- The cost assumption;

the cost should be equal to all destination nodes. eg: if one load costs 9, all loads to that destination should be 9

  - The cost of distributing units from any particular source to any particular destination is directly proportional to the number of units distributed.

Property

- **The feasible solutions property:** A transportation problem will have feasible solutions if and only if

$$\sum_{i=1}^{m} s_i = \sum_{j=1}^{n} d_j.$$

- **Integer solutions property:** For transportation problems where every s_i and d_j have an integer value, all the basic variables (allocations) in *every* basic feasible (BF) solution (including an optimal one) also have *integer* values.

LP Formulation for Transportation Problem

$$\text{Minimise}: \quad Z = \sum_{i=1}^{n} \sum_{j=1}^{m} c_{ij} x_{ij}$$

supply node equation

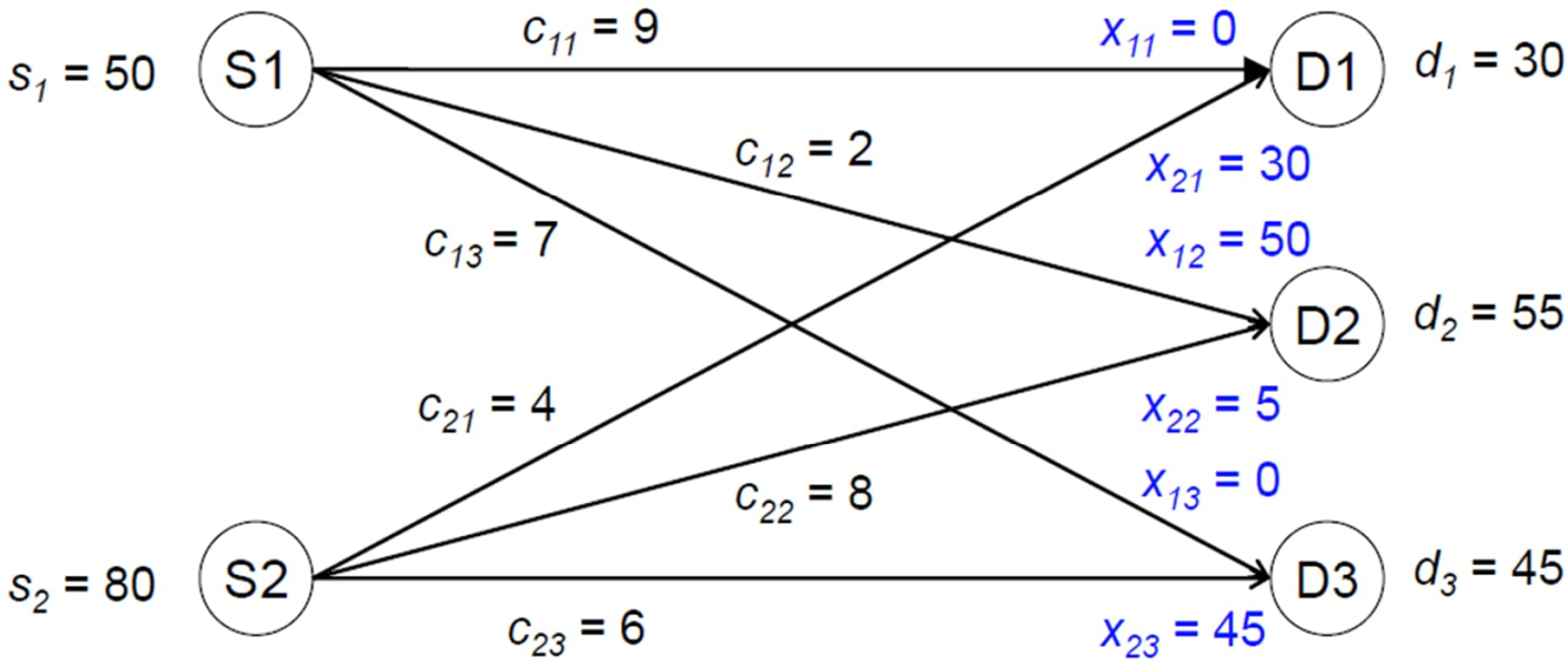
$$\text{Subject to}: \quad \sum_{j=1}^{m} x_{ij} = s_i \quad \text{for } i = 1,2 \ldots n$$

demand node equation

$$\sum_{i=1}^{n} x_{ij} = d_j \quad \text{for } j = 1,2 \ldots m$$

$$x_{ij} \geq 0 \quad \text{for all } i \text{ and } j$$

This model of the transportation problem assumes that the problem is balanced.

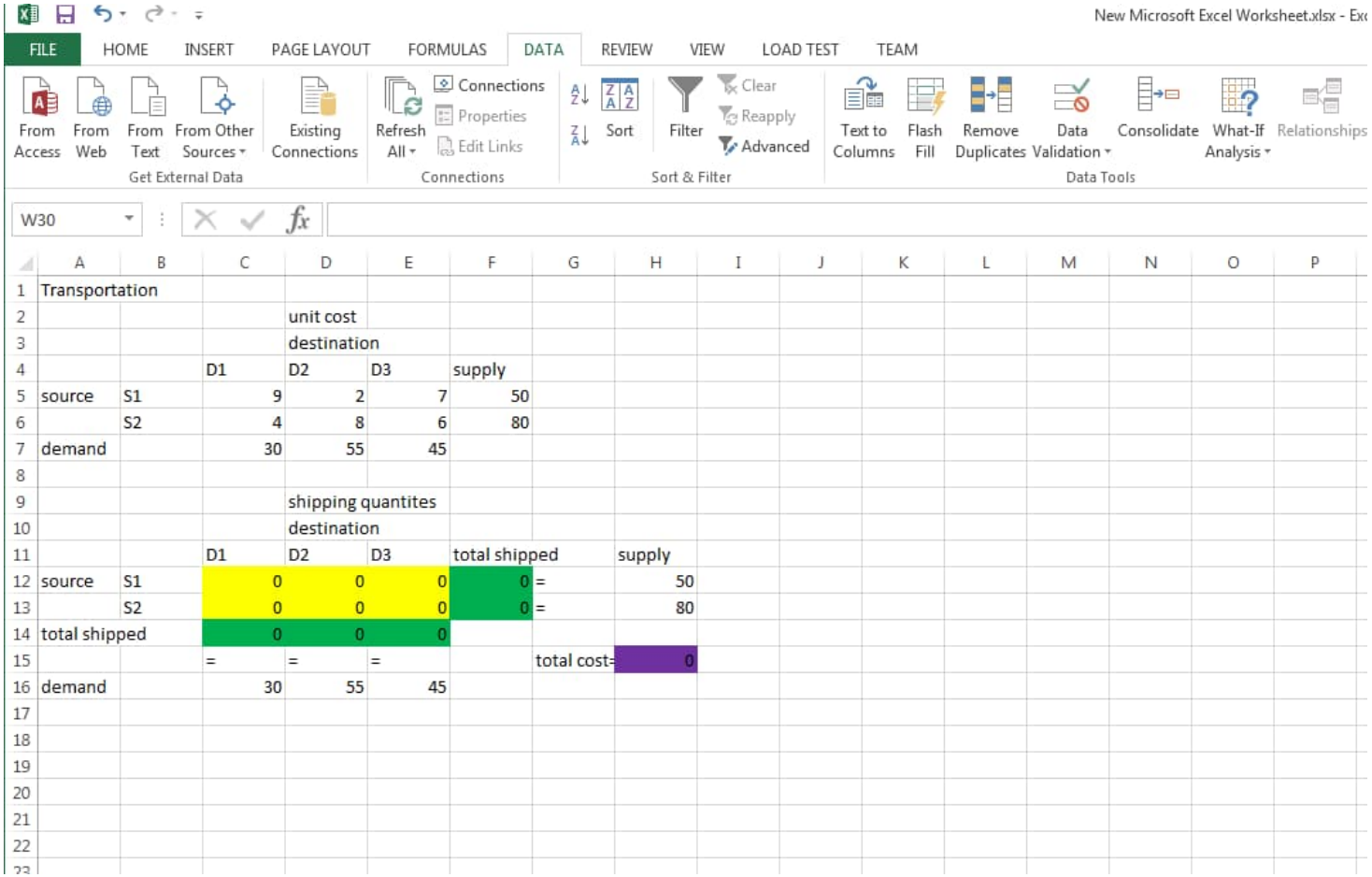

$s_1 = 50$
S1
$c_{11} = 9$
$x_{11} = 0$
D1
$d_1 = 30$
$c_{12} = 2$
$x_{21} = 30$
$c_{13} = 7$
$x_{12} = 50$
D2
$d_2 = 55$
$c_{21} = 4$
$x_{22} = 5$
$x_{13} = 0$
$c_{22} = 8$
$s_2 = 80$
S2
D3
$d_3 = 45$
$c_{23} = 6$
$x_{23} = 45$

$$
\begin{aligned}
&\text{Minimise}: \quad Z = 9x_{11} + 2x_{12} + 7x_{13} + 4x_{21} + 8x_{22} + 6x_{23} \\
&\text{Subject to}: \quad x_{11} + x_{12} + x_{13} = 50 && (1) \\
&\qquad\qquad\quad x_{21} + x_{22} + x_{23} = 80 && (2) \\
&\qquad\qquad\quad x_{11} + x_{21} = 30 && (3) \\
&\qquad\qquad\quad x_{12} + x_{22} = 55 && (4) \\
&\qquad\qquad\quad x_{13} + x_{23} = 45 && (5) \\
&\qquad\qquad\quad x_{ij} \geq 0 \quad \text{for all } i \text{ and } j && (6)
\end{aligned}
$$

Minimum Transportation Cost: 530

TABLE Parameter table for the transportation problem

		Cost per Unit Distributed				
		Destination				
		1	**2**	⋯	***n***	**Supply**
Source	1	c_{11}	c_{12}	⋯	c_{1n}	s_1
	2	c_{21}	c_{22}	⋯	c_{2n}	s_2
	⋮	⋯	⋯	⋯	⋯	⋮
	m	c_{m1}	c_{m2}	⋯	c_{mn}	s_m
Demand		d_1	d_2	⋯	d_n	



	A	B	C	D	E	F	G	H
1	Transportation							
2				unit cost				
3				destination				
4			D1	D2	D3	supply		
5	source	S1	9	2	7	50		
6		S2	4	8	6	80		
7	demand		30	55	45			
8								
9				shipping quantites				
10				destination				
11			D1	D2	D3	total shipped		supply
12	source	S1	0	0	0	0	=	50
13		S2	0	0	0	0	=	80
14	total shipped		0	0	0			
15			=	=	=		total cost=	0
16	demand		30	55	45			

Special algorithm for the transportation problem

- Excel solver applies Simplex LP. Thus, it is not necessary to do the reformulation to maintain the $\sum_{i=1}^{m} s_i = \sum_{j=1}^{n} d_j$. since the Simplex method can solve the original model.
- For large transportation problem, a streamlined version of the Simplex is available, thus it is important to reformulate to achieve $\sum_{i=1}^{m} s_i = \sum_{j=1}^{n} d_j$.

Assignment problem

- A special type of LP where assignees are being assigned to perform tasks.

  In this type of problem, the x values would either be 0 or 1
- e.g. assign people to jobs. Assignees can be machines, vehicles, plants, or time slots, etc.
- Problem Assumptions:
  - Number of assignees=number of tasks;
  - Each assignee is to be assigned to exactly *one* task;
  - Each task is to be performed by exactly *one* assignee;
  - There is a cost associated with assignee performing task;
  - The objective is to determine how n assignments should be made to minimise the total cost.

Assignment problem-a prototype example

The Assignment Problem

There is a set of N tasks and a set of N workers with a certain cost c_{ij} for assigning each task to each worker. Each task must be assigned to exactly one worker and each worker must undertake exactly one task. The problem is to assign all the tasks to the workers so that the total cost is minimised.

Minimise : $Z = \sum_{i=1}^{N} \sum_{j=1}^{N} c_{ij} x_{ij}$

Subject to: $\sum_{j=1}^{N} x_{ij} = 1 \quad \text{for } i = 1,2,\ldots N \quad (1)$

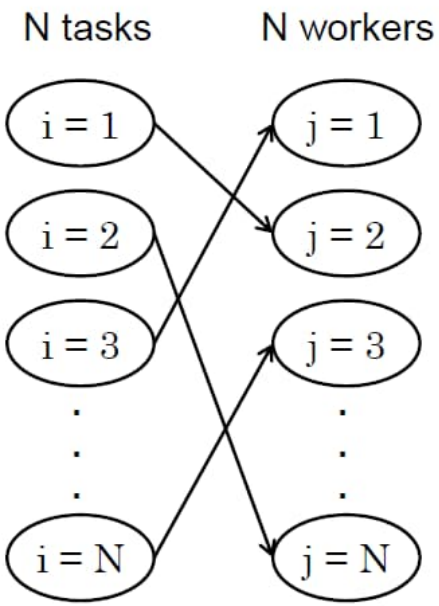
$\sum_{i=1}^{N} x_{ij} = 1 \quad \text{for } j = 1,2,\ldots N \quad (2)$

$x_{ij} = 1$ if task i is assigned to worker j, 0 otherwise

Property

- The problem model presents an integer constraint,
- Assignment problem is a special type of transportation problem.
- we *can* delete the integer restriction due to the *integer solutions property* on the transportation problem model.

Sketch of the Assignment Problem



Given N there are :

- N^2 decision variables
- N^2 assignment costs
- $N!$ possible feasible solutions
- 2^{N^2} possible solutions (feasible + infeasible)

For a problem with $N = 3$:

Minimise : $Z = c_{11}x_{11} + c_{12}x_{12} + c_{13}x_{13} +$
$c_{21}x_{21} + c_{22}x_{22} + c_{23}x_{23} +$
$c_{31}x_{31} + c_{32}x_{32} + c_{33}x_{33}$

Subject to :

$$x_{11} + x_{12} + x_{13} = 1 \quad (1a)$$
$$x_{21} + x_{22} + x_{23} = 1 \quad (1b)$$
$$x_{31} + x_{32} + x_{33} = 1 \quad (1c)$$
$$x_{11} + x_{21} + x_{31} = 1 \quad (2a)$$
$$x_{12} + x_{22} + x_{32} = 1 \quad (2b)$$
$$x_{13} + x_{23} + x_{33} = 1 \quad (2c)$$
$$x_{ij} \in \{0,1\}$$

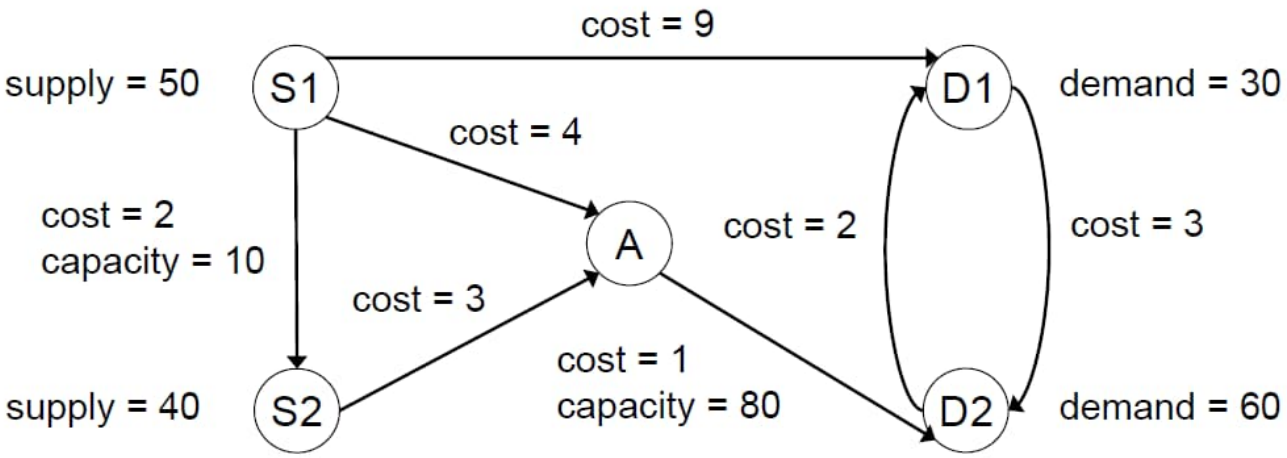
Minimum cost flow problem

- Holds a central position among network optimisation models, both because it encompasses a broad class of applications and because it can be solved efficiently;
- Description of the problem:
  - The network is a *directed* and *connected* network;
  - *At least one* of the nodes is a *supply node*;
  - *At least one* of the other nodes is *demand node*;
  - All the remaining nodes are *transhipment nodes*;
  - Flow through the arc is allowed only in the direction indicated by the arrowhead. The maximum amount of flow is given by the *capacity* of that arc.
  - The network has enough arc with sufficient capacity to enable all the flow generated at *supply nodes* to reach all the *demand nodes*;
  - The cost of the flow through each arc *is proportional* to the amount of that flow and the unit cost is known;
  - The objective is to minimise the total cost of sending the available supply through the network to satisfy the given demand.

Minimum cost flow-A prototype example

Minimum Cost Flow Problem

Consider a directed network with a set of nodes, at least one node supplies flow, at least one node demands flow and a set of transhipment nodes. A given cost of flow and non-negative maximum flow capacity are associated to each arc. The objective is to distribute the entire flow supply through the network from the sources to the destinations such as to minimise the total flow cost. Linear Programming can solve the transportation problem very efficiently.



Consider a directed and connected network where the n nodes include at least one supply node and at least one demand node. The decision variables are

x_{ij} = flow through arc $i \rightarrow j$,

and the given information includes

c_{ij} = cost per unit flow through arc $i \rightarrow j$,
u_{ij} = arc capacity for arc $i \rightarrow j$,
b_i = net flow generated at node i.

The value of b_i depends on the nature of node i, where

$b_i > 0$ if node i is a supply node,
$b_i < 0$ if node i is a demand node,
$b_i = 0$ if node i is a transshipment node.

LP Formulation for Minimum Cost Flow Problem

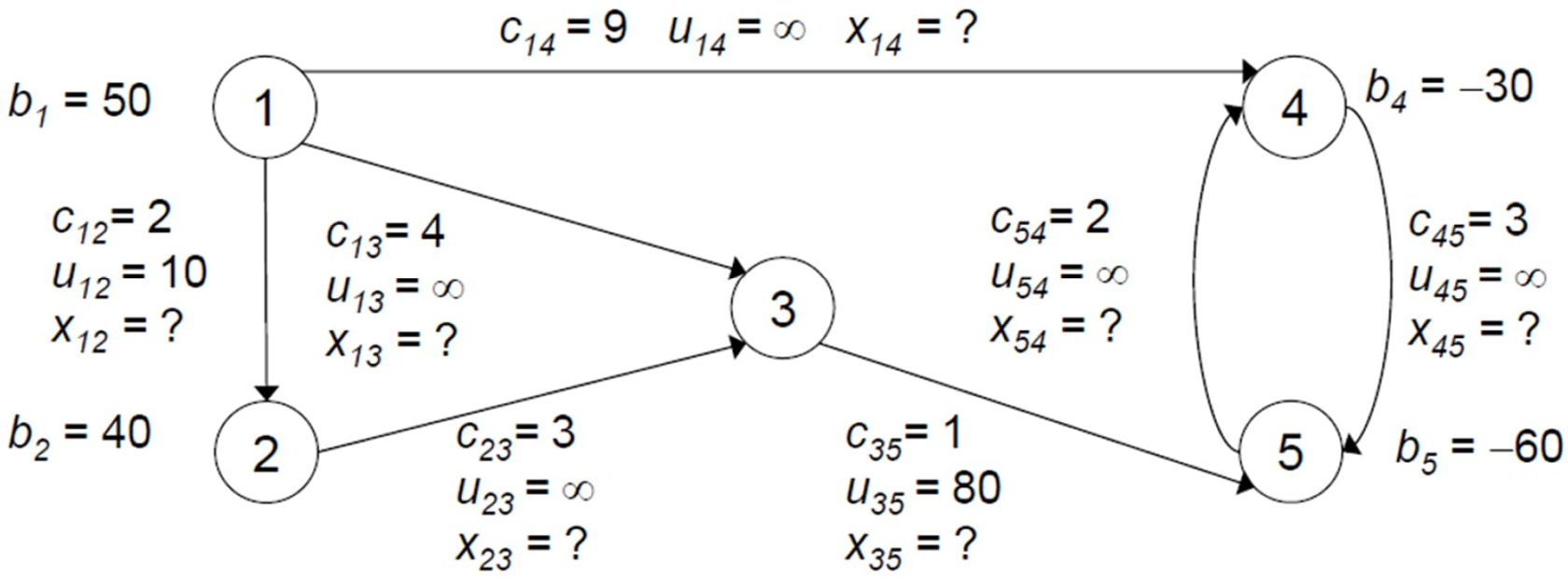
$$\text{Minimise}: \quad Z = \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij}$$

$$\text{Subject to}: \quad \sum_{j=1}^{n} x_{ij} - \sum_{j=1}^{n} x_{ji} = b_i \quad \text{for } i = 1,2\ldots n$$

$$0 \le x_{ij} \le u_{ij} \quad \text{for each arc } i \rightarrow j$$

This model of the minimum cost flow problem is a more general case of the transportation problem.

Minimum cost flow



Minimum cost flow

$$\text{Minimise}: \quad Z = 2x_{12} + 4x_{13} + 9x_{14} + 3x_{23} + x_{35} + 3x_{45} + 2x_{54}$$

$$\text{Subject to}: \quad x_{12} + x_{13} + x_{14} = 50 \qquad (1)$$

$$x_{23} - x_{12} = 40 \qquad (2)$$

$$x_{35} - x_{13} - x_{23} = 0 \qquad (3)$$

$$x_{45} - x_{14} - x_{54} = -30 \qquad (4)$$

$$x_{54} - x_{35} - x_{45} = -60 \qquad (5)$$

$$0 \le x_{12} \le 10\,,\ 0 \le x_{35} \le 80 \ \text{ and } \ x_{ij} \ge 0 \qquad (6)$$

Network models

- Minimum cost flow, a unified approach:
  - Transportation problem; (no transhipment nodes, no capacity limits)
  - Assignment problem; (a special type of transportation problem)
  - Maximum flow problem;
  - Shortest-path problem;

Reading

- Paper: Application of networks

Questions?